

IT223 - Web Systems and Technologies

DREAMHOME

IT2R6

Group 6:

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PROBLEM STATEMENT

- Many clients struggle to find properties matching their preferences quickly.
- Staff have difficulty tracking client requests and managing properties efficiently.

PURPOSE

DreamHome aims to streamline property search, client registration, and staff assignment.

TARGET USERS & INTENDED USE

- **Target Users:**
 - Clients searching for rental properties
 - Staff managing client inquiries and property listings
- **Intended Use:**
 - Allow clients to view properties based on preferences
 - Allow staff to track client requests, assign staff, and update statuses

SYSTEM FEATURES, MODULES, USER ROLES

- **Features:**

- Client registration and property search
- Staff assignment and management
- Viewing requests and inspections

- **Modules:**

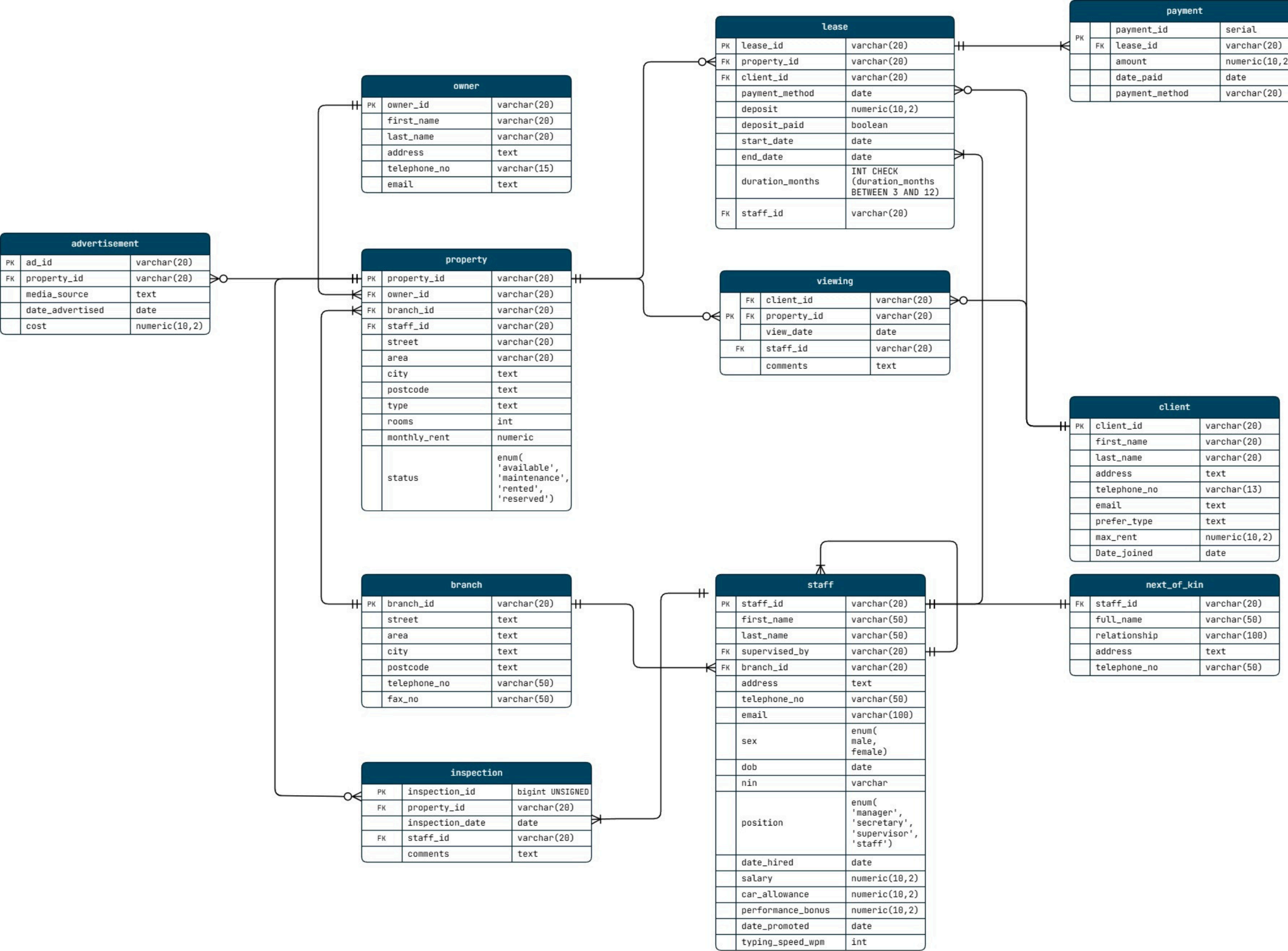
- Client Module
- Staff Module
- Property Module
- Branch Module
- Owner Module

- **User Roles:**

- Client
- Staff
- Admin

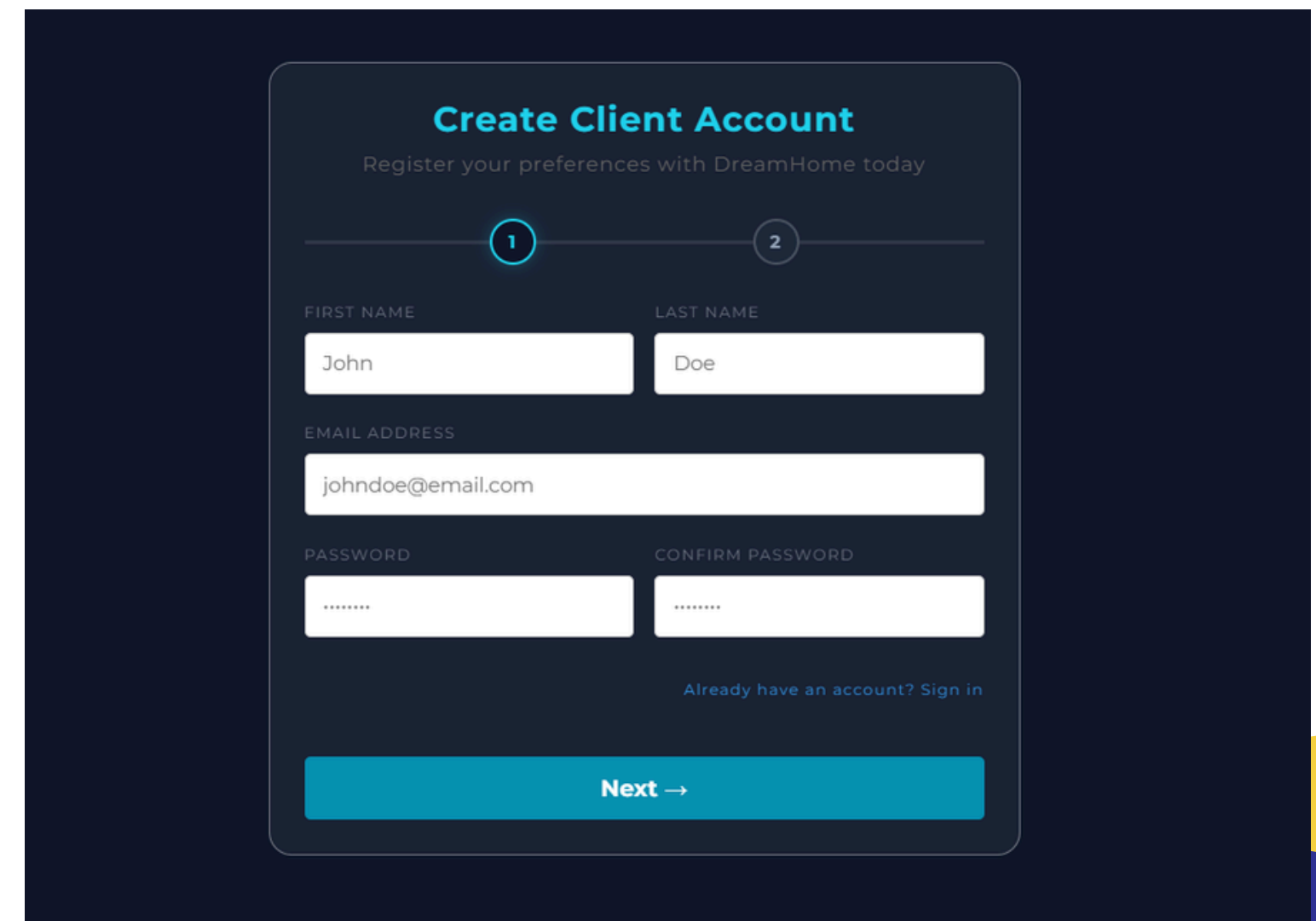
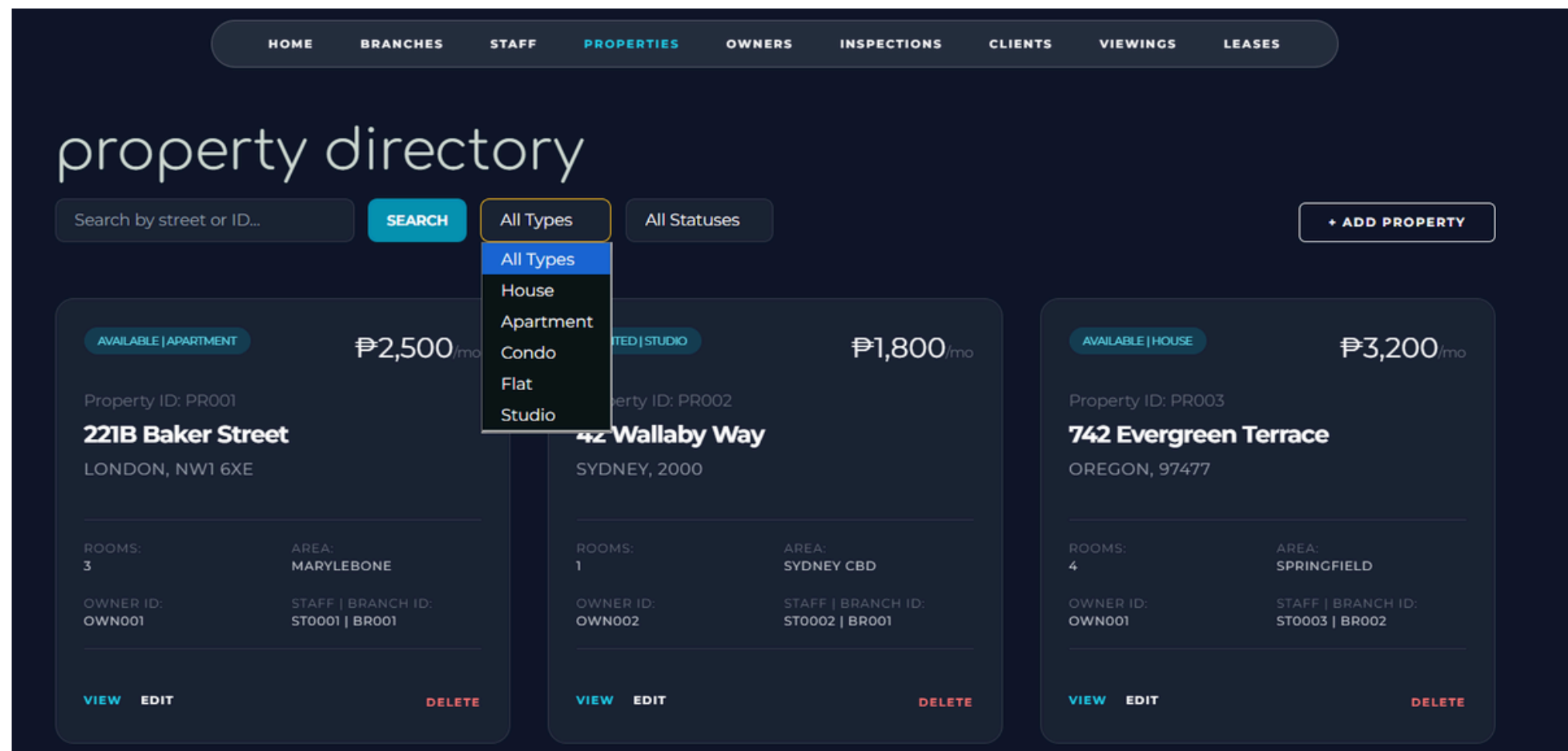
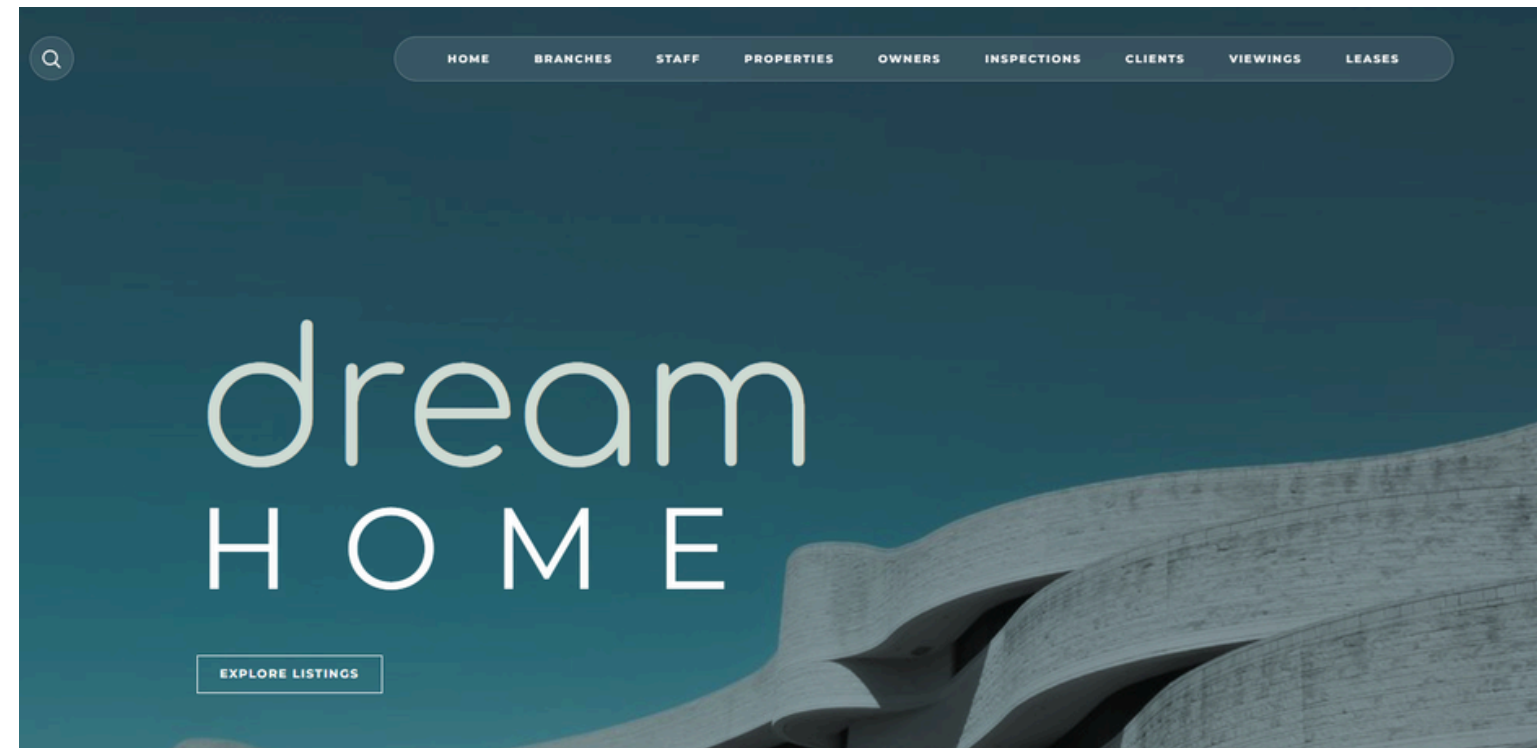
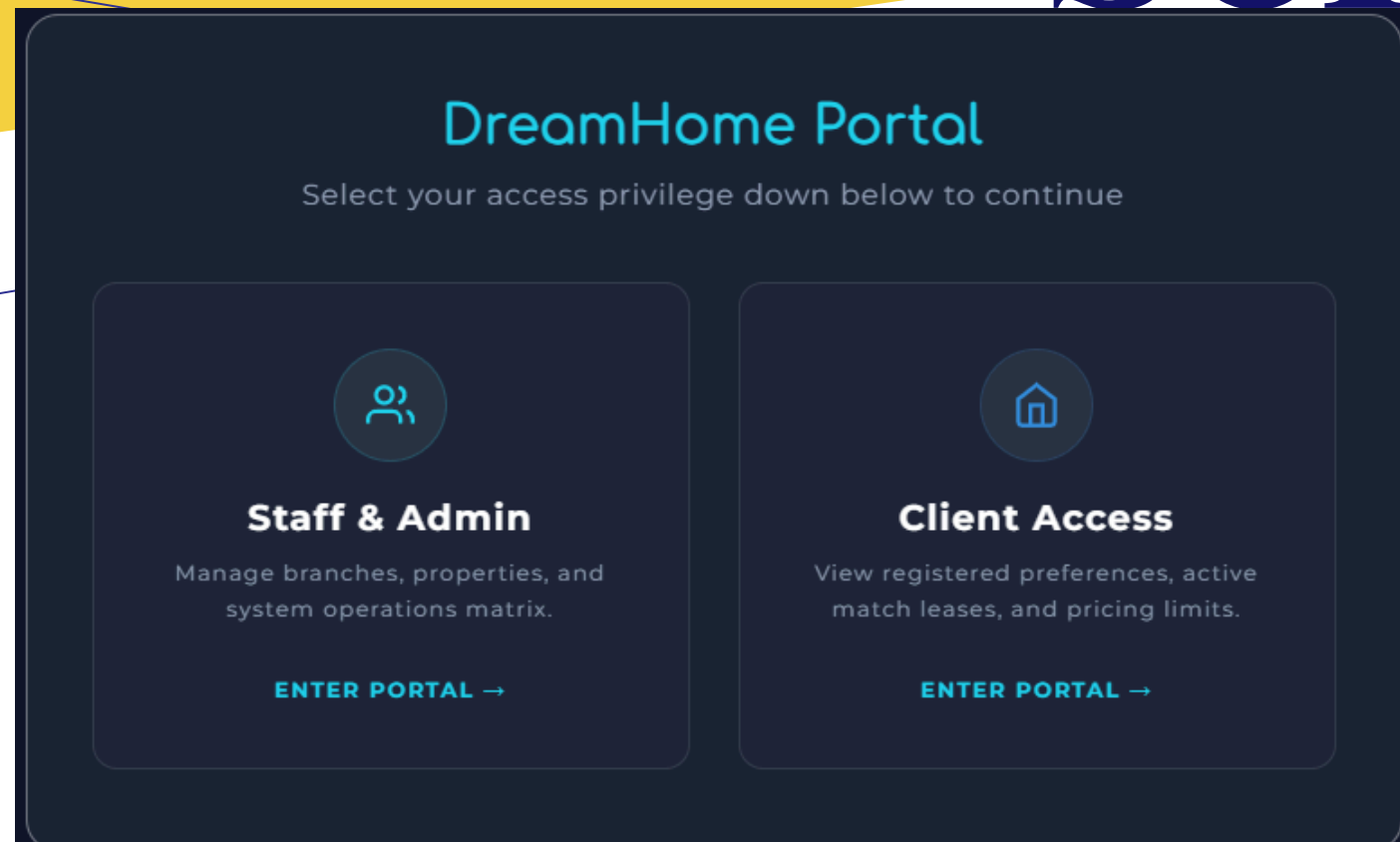
TECHNOLOGIES USED

- **Frontend:**
 - HTML, CSS, JavaScript
- **Backend:**
 - Laravel (PHP)
- **Database:**
 - PostgreSQL
- **Other Tools:**
 - Git, VS Code, Flutter (mobile app), Thunder Client for testing APIs



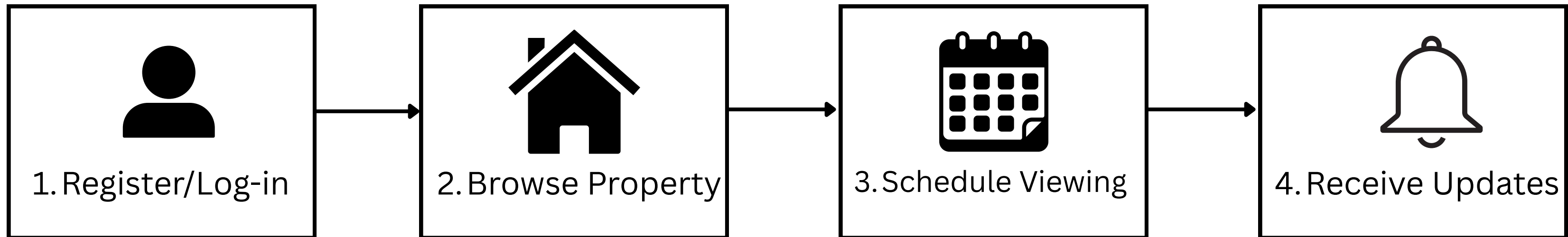
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SCREENSHOTS

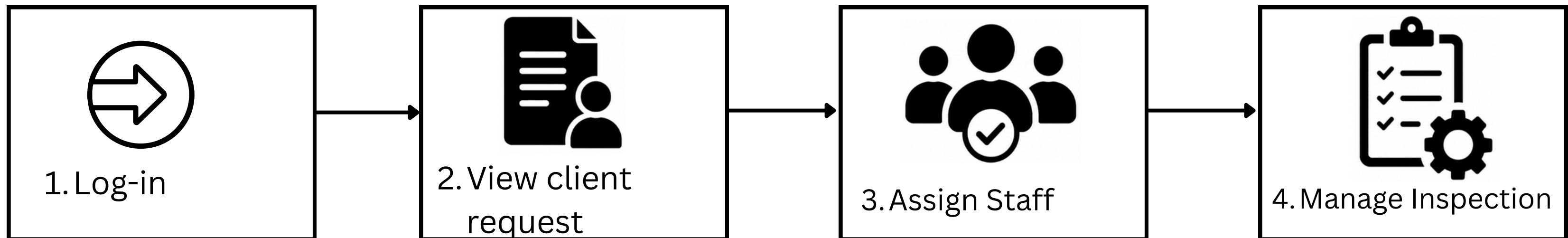


SYSTEM WORKFLOW

Client Flow Chart



Staff Flow Chart



CHALLENGES & SOLUTIONS

Challenges	Solution
Laravel and Flutter connection	Used REST API endpoints and configured Flutter base URL for emulator testing.
Authentication for protected routes	Stored the returned client token and sent it in Authorization: Bearer token headers.
Validation errors	Used Laravel validation and displayed messages in Flutter snackbars or Thunder Client responses
Property filtering	Compared client maximum rent with property monthly rent before displaying properties in the mobile app.
Viewing workflow	Created Pending viewing requests first, then allowed staff/admin side to schedule and update status.

CONCLUSION, LIMITATIONS, IMPROVEMENTS

The DreamHome Rental Management System successfully demonstrates a Laravel web application integrated with a Flutter mobile application through REST API endpoints. The system allows staff and administrators to manage rental records, while clients can register, log in, browse properties, request viewings, and check their viewing status.

The project meets the main requirements, including an existing Laravel web system, Flutter mobile application, login authentication, REST API integration, multiple API endpoints, and working mobile features.

CONCLUSION, LIMITATIONS, IMPROVEMENTS

Some limitations include the need for improved deployment, better UI consistency, more advanced search and filtering, notification features, and stronger role-based access control.

Possible improvements include adding email or mobile notifications, improving the dashboard design, adding advanced property recommendations, deploying the system online, and improving security and validation before production use.